

METHODOLOGY

Problem Statement

Developing computer vision solutions for the FMCG and retail sectors, such as shelf-auditing and planogram compliance—annotating tens of thousands of images for every new product SKU is frequently impractical and time-consuming. This project shows a pipeline that achieves 96% classification accuracy using 100 labeled images.

Dataset

I had used the **Food101 dataset** which is an academic dataset built into PyTorch. It was easy to download this dataset as it does not require any API keys or manual setup.

I had taken 20 images per class across 5 categories ($20 \times 5 = 100$ images) to create a real world low label FMCG.

Classes selected:

1. chocolate_mouuse
2. waffles
3. hot_and_sour_soup
4. spaghetti_carbonara
5. edamame

Since I had to restrict myself to 100 images only therefore all the 5 classes selected were very different in appearance hence easy to train and classify.

Data Splits

Train	100 images
Test	50 images
Validation	150 images

Strategies To Reduce Labeling Time

1. Transfer Learning

Instead of training from scratch as it requires thousands of images I used the EfficientNet-B2 model pretrained on ImageNet-1k. This model knows how to detect edges, shapes and textures so I was only required to teach it the final classification step for the 5 product classes. It saved my labeling time.

2. Selective Layer Freezing

With only 100 training images, training on all millions of parameters caused overfitting so I froze all the layers except the last convolutional block & classification head. The frozen layer retained powerful ImageNet representation without being corrupted by small data.

3. Repeated Sampling

Each of the 100 images were shown to the model 15 times per epoch with a different augmentation every time, which created effective training samples per epoch from just 100 real

images.

Augmentation: flip, rotation, random erasing, random crop

4. Mixup Augmentation

Mixup blends 2 training images together during each batch. This helps the model to learn a smooth transition between classes.

Model Architecture

Model	Efficient-B2
Optimizer	AdamW
Learning Rate	1e-3
LR Schedule	Cosine annealing
Batch size	16
Max epoch	50
Repeat factor	15x

Results

Pretrained model with no fine-tuning

~45% accuracy

Fine tuned with no augmentation

~72% accuracy

Fine tuned + heavy augmentation + repeat sampling

~80% accuracy

Fine tuned + heavy augmentation + repeat sampling + Mixup

~96% accuracy

Final Test Accuracy

~98%

Training Curve



Confusion Matrix

